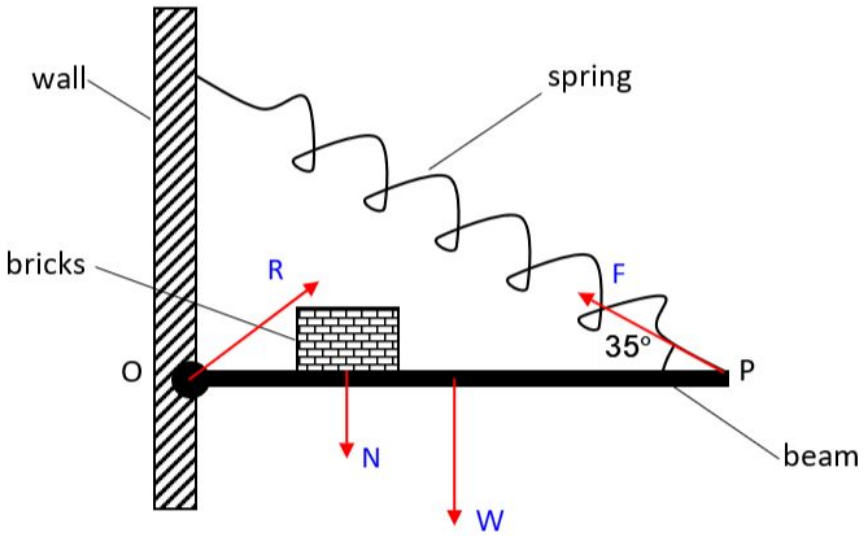
Qns	Answer	Marks
1(a)	Moment of a force is the product of the force and its perpendicular distance to the pivot.	B1
1(b)(i)	$\cos 35° = \frac{6.8}{y}$ $y = \frac{6.8}{\cos 35°} = 8.3$ m extension $x = 8.3 - 7.0$ $= 1.3$ m	M1 A0
1(b)(ii)	 W: weight of beam (or gravitational force by earth on beam) N: normal contact/reaction force by bricks on beam F: force by spring on beam (or spring force / tension / elastic force) (EITHER) R: contact/reaction force by wall/hinge on beam (OR) $R_{vertical}$: vertical component of contact/reaction force by wall/hinge on beam $R_{horizontal}$: horizontal component of contact/reaction force by wall/hinge on beam (Alternatively): normal contact/reaction force by wall/hinge on beam Correct position and direction of all forces Length of arrow of W longer than N	B1 B1

Qns	Answer	Marks
1(b)(iii)	For rotational equilibrium, taking moment about the pivot O, sum of clockwise moment = sum of anticlockwise moment $N(1.8)+W(3.4)=(F\sin 35°)(6.8)$ $(120)(9.81)(1.8)+(190)(9.81)(3.4)=(F\sin 35°)(6.8)$ $F=2168$ N $F=2200$ N or $F=2170$ N $k=\frac{F}{x}=\frac{2200}{1.3}$ or $k=\frac{F}{x}=\frac{2170}{1.3}$ $=1692$ N m^{-1} $=1669$ N m^{-1} $=1700$ N m^{-1} (2 s.f.) $=1700$ N m^{-1} (2 s.f.) Accept both 2 s.f. and 3 s.f. **Note: accept 1670 N m^{-1}, 1690 N m^{-1} and 1700 N m^{-1}.**	**M1** **A1**
1(b)(iv)	$E_P=\frac{1}{2}kx^2$ $=\frac{1}{2}(1700)(1.3)^2$ $=1437$ $=1400$ J (2 s.f.) Note: Can also use $E_P=\frac{1}{2}Fx$ or $E_P=\frac{1}{2}\left(\frac{F^2}{k}\right)$ Accept both 2 s.f. and 3 s.f. Allow ECF. **Note: accept 1400 J, 1410 J, 1430 J and 1440 J**	**C1** **A1**
1(b)(v)	As the pile of bricks is moved towards the pivot O, the clockwise moment about the pivot (due to the bricks) decreases (as the perpendicular distance from pivot to line of action of the force decreases). (The total clockwise moment decreases). A lower anti-clockwise moment is needed to maintain equilibrium. Therefore, the force exerted by the string (on the beam) decreases.	**B1** **B1**